

Executive Summary – FNP Sales Analysis Dashboard

Project Overview

The **FNP Sales Analysis Dashboard** is an end-to-end Business Intelligence project developed in **Microsoft Excel** to analyze sales performance and customer purchasing behavior for **Ferns N Petals (FNP)**. The dashboard transforms raw transactional sales data into an interactive and visually appealing reporting solution that enables business users to monitor key performance indicators (KPIs), identify sales trends, evaluate product performance, and support strategic decision-making.

The project analyzes **1,000 customer orders** across multiple occasions, product categories, cities, and time periods. By leveraging advanced Excel features such as **Power Query Editor, Power Pivot, Data Modeling, DAX Measures, Pivot Tables, Pivot Charts, Slicers, Timelines**, and **Dynamic KPI Cards**, the dashboard provides a comprehensive view of business performance while allowing users to interactively filter and explore the data.

This project demonstrates how Microsoft Excel can be utilized as a powerful Business Intelligence and Data Analytics tool to convert raw business data into meaningful insights that help management improve sales strategies, marketing effectiveness, inventory planning, and customer experience.

Business Problem

Ferns N Petals receives thousands of customer orders across different gifting occasions, cities, product categories, and delivery dates. However, analyzing such large volumes of transactional data manually makes it difficult for management to identify revenue patterns, customer behavior, seasonal demand, and product performance.

Without a centralized reporting system, business stakeholders face challenges in:

- Tracking overall sales performance.
- Identifying the most profitable occasions.
- Understanding customer purchasing patterns.
- Measuring category-wise revenue contribution.
- Monitoring monthly sales trends.
- Identifying top-selling products.
- Evaluating regional market performance.
- Making quick, data-driven business decisions.

To address these challenges, an interactive Excel dashboard was developed to consolidate data, automate calculations, and provide actionable business insights through intuitive visualizations.

Project Objectives

The primary objectives of this project were to:

- Monitor overall business performance using real-time KPIs.
- Analyze revenue generated across different gifting occasions.
- Identify the highest revenue-generating product categories.
- Understand customer purchasing behavior based on order timing.
- Evaluate monthly and seasonal revenue trends.
- Identify top-performing products based on sales revenue.
- Analyze city-wise order distribution.
- Measure average customer spending.
- Monitor average order delivery time.
- Build an interactive management dashboard using Excel Business Intelligence tools.

Data Preparation & Modeling

Before performing analysis, the raw dataset underwent extensive preprocessing to ensure data quality and consistency.

Power Query Editor

Power Query Editor was used for:

- Importing raw sales data.
- Cleaning inconsistent records.
- Removing duplicate entries.
- Handling missing values.
- Standardizing date and text formats.
- Transforming columns for analysis.
- Creating a structured dataset for reporting.

Using Power Query significantly reduced manual effort while ensuring data accuracy and refreshability.

Data Modeling using Power Pivot

Power Pivot was used to build a relational data model by connecting multiple tables through appropriate relationships. This approach enabled efficient data aggregation and improved dashboard performance when working with large datasets.

The data model served as the foundation for advanced calculations and interactive reporting.

DAX Measures

Several **Data Analysis Expressions (DAX)** measures were created to calculate dynamic business metrics and KPIs.

Examples include:

- Total Orders
- Total Revenue
- Average Customer Spend
- Average Order Delivery Time
- Revenue by Occasion
- Revenue by Product Category
- Monthly Revenue
- Top Product Revenue

Using DAX ensured that all KPIs automatically update whenever filters or slicers are applied.

Dashboard Components

The dashboard includes multiple analytical sections designed to answer important business questions.

1. KPI Summary Cards

The dashboard displays high-level business metrics including:

- Total Orders
- Total Revenue
- Average Customer Spending
- Average Order Delivery Time

These KPIs provide an instant overview of overall business performance.

2. Revenue Analysis by Occasion

This visualization compares revenue generated across different gifting occasions.

Key Findings

- Anniversary generated the highest revenue.
- Raksha Bandhan contributed significantly to overall sales.
- Holi maintained strong seasonal performance.
- Birthday sales remained relatively stable.
- Valentine's Day generated comparatively lower revenue.
- Diwali showed the lowest revenue among major occasions.

Business Insight

The analysis indicates that Anniversary and Raksha Bandhan represent the company's strongest revenue-generating events. Increasing marketing investments, promotional campaigns, and inventory availability during these occasions can significantly improve annual sales performance.

3. Revenue by Product Category

The dashboard evaluates revenue contribution across different product categories including:

- Cakes
- Soft Toys
- Sweets
- Plants
- Rakhi
- Mugs
- Gift Accessories

Key Findings

- One product category contributes nearly 30% of the total revenue.
- Some categories consistently outperform others.

- Lower-performing categories present opportunities for improvement.

Business Recommendations

- Introduce combo offers.
- Cross-sell complementary products.
- Create festival bundles.
- Offer personalized recommendations.
- Launch targeted promotional discounts.

4. Revenue by Order Time

The hourly sales analysis helps understand customer purchasing behavior.

Observations

- Customer orders fluctuate throughout the day.
- Multiple revenue peaks indicate purchasing activity across different time slots.
- Certain hours consistently generate higher sales.

Business Recommendation

Businesses should schedule:

- Flash Sales
- Push Notifications
- Email Campaigns
- Social Media Promotions
- Paid Advertisements

during peak ordering hours to maximize customer engagement and conversions.

5. Monthly Revenue Trend

The monthly trend analysis identifies seasonal sales patterns throughout the year.

Key Findings

- February recorded the highest revenue.

- Revenue declined after the seasonal peak.
- Sales stabilized during subsequent months.
- Year-end months maintained relatively consistent performance.

Business Recommendation

Before high-demand months, the company should increase:

- Inventory levels
- Delivery workforce
- Marketing budget
- Customer support capacity

to ensure smooth operations during seasonal demand.

6. Top Products by Revenue

The dashboard highlights the five highest revenue-generating products.

Findings

The leading products include:

- Magnum Set
- Premium Gift Packs
- Hamper Collections
- Decorative Gift Boxes
- Gift Bundles

Business Recommendation

These products should receive:

- Priority inventory allocation
- Homepage placement
- Premium advertising
- Bundle recommendations
- Personalized marketing campaigns

to maximize profitability.

7. Top Cities by Orders

Regional analysis identifies cities generating the highest order volumes.

Key Findings

High-performing cities include:

- Dhanbad
- Imphal
- Kavali

Business Insight

These cities represent strong customer markets and should be prioritized through:

- Localized marketing campaigns
- Faster delivery services
- Regional promotional offers
- Customer loyalty programs

to further increase market share.

Dashboard Features

The dashboard incorporates several interactive Excel features to enhance usability and decision-making.

Interactive Features

- Dynamic KPI Cards
- Power Query Data Transformation
- Power Pivot Data Modeling
- DAX Measures
- Pivot Tables
- Pivot Charts

- Interactive Slicers
- Timeline Filters
- Occasion Filters
- Order Date Filters
- Delivery Date Filters
- Dynamic Visualizations
- Automated Calculations
- Business-Friendly Layout

These features allow users to filter the dashboard instantly and explore insights from multiple business perspectives.

Business Recommendations

Based on the analysis, the following strategic recommendations are proposed:

1. Increase Marketing Investment

Focus advertising campaigns around:

- Anniversary
- Raksha Bandhan
- Holi

to capitalize on peak seasonal demand.

2. Improve Underperforming Categories

Increase sales through:

- Combo Deals
- Festival Packages
- Personalized Recommendations
- Cross-selling Strategies
- Discount Campaigns

3. Optimize Delivery Operations

Reduce the average delivery time from approximately **5.7 days** to below **4 days** to improve customer satisfaction and encourage repeat purchases.

4. Utilize Peak Ordering Hours

Launch promotions, email campaigns, and digital advertisements during high-conversion time slots identified in the dashboard.

5. Strengthen Regional Marketing

Expand marketing efforts in top-performing cities while identifying growth opportunities in lower-performing regions.

6. Inventory Planning

Maintain higher inventory levels for best-selling products before major gifting occasions to prevent stock shortages and maximize revenue.

Tools & Technologies Used

- Microsoft Excel
- Power Query Editor
- Power Pivot
- Data Modeling
- DAX (Data Analysis Expressions)
- Pivot Tables
- Pivot Charts
- Slicers
- Timelines
- KPI Cards
- Data Cleaning
- Data Transformation
- Data Visualization
- Business Intelligence Reporting
- Sales Analytics
- Dashboard Design

- Interactive Reporting

Key Skills Demonstrated

- Data Cleaning & Transformation
- Data Modeling
- Business Intelligence
- Dashboard Development
- KPI Design
- DAX Calculations
- Sales Performance Analysis
- Customer Behavior Analysis
- Data Visualization
- Interactive Reporting
- Business Analytics
- Decision Support
- Microsoft Excel Advanced Features

Overall Conclusion

The **FNP Sales Analysis Dashboard** demonstrates how Microsoft Excel can be transformed into a powerful Business Intelligence solution capable of analyzing large-scale transactional data and generating meaningful business insights. By integrating **Power Query** for data preparation, **Power Pivot** for data modeling, **DAX** for dynamic calculations, and interactive visualizations using **Pivot Tables, Pivot Charts, Slicers, and Timelines**, the dashboard provides stakeholders with a comprehensive and user-friendly reporting experience.

The analysis successfully identifies high-performing occasions, top-selling products, customer spending behavior, peak ordering hours, monthly sales trends, and regional demand patterns. These insights enable management to make informed decisions related to marketing strategies, inventory planning, customer engagement, operational efficiency, and revenue growth.

Overall, this project showcases practical expertise in **Excel-based Business Intelligence**, advanced analytical techniques, and interactive dashboard development, demonstrating the ability to convert raw business data into actionable insights that support strategic planning and organizational decision-making.